

LAB REPORT: ASSIGNMENT 03

Topic: Sorting data Using Python & R

- **Course Name:** Introduction To Data Science
- **Course Code :** ADS 201
- **Course Instructor:** Dr. Muhammad Abul Hasan
- **Student Name:** Momota Sardar Maya
- **Student ID:** 252035002
- **Date:** 28th July 2026

1. Objective

The objective of this assignment is to sort a dataset in ascending order based on a specific column using both Python and R.

2. Dataset Overview

- **Dataset Name:** Space_Corrected.csv
- **Description:** A dataset containing historical space mission information such as company name, launch location, mission details, rocket status, mission status, and other related information.

Tasks Performed

1. Loaded the CSV dataset.
2. Created two dummy score columns (Score1 and Score2).
3. Calculated a new column named Average Score.
4. Sorted the dataset in ascending order based on the Average Score column.
5. Displayed the first few rows of the sorted dataset.

3. Part A: Implementation in Python

Steps Taken:

1. Loaded the dataset using the pandas library.
2. Created the Average Score column.
3. Sorted the dataset using the `sort_values()` function with ascending order.
4. Displayed the first rows of the sorted dataset.

Python Code:

```

1 import pandas as pd
2
3 # Load dataset
4 df = pd.read_csv("Space_Corrected.csv")
5
6 # Create dummy score columns
7 df["Score1"] = (df.index % 21) + 70
8 df["Score2"] = (df.index % 16) + 75
9
10 # Create Average Score
11 df["Average Score"] = df[["Score1", "Score2"]].mean(axis=1)
12
13 # Sort data by Average Score (Ascending)
14 sorted_df = df.sort_values(by="Average Score", ascending=True)
15
16 # Display first 10 rows
17 print(sorted_df[["Score1", "Score2", "Average Score"]].head(10))

```

Output & Observations:

The dataset was successfully sorted in ascending order based on **Average Score**. The output shows the lowest calculated score (72.5) displayed at the top.

Run assignment_3

	Score1	Score2	Average Score
0	70	75	72.5
672	70	75	72.5
1680	70	75	72.5
3024	70	75	72.5
3696	70	75	72.5
1008	70	75	72.5
4032	70	75	72.5

4. Part B: Implementation in R

Steps Taken:

1. Loaded the dataset using the `read.csv()` function.
2. Created two dummy score columns (**Score1** and **Score2**).
3. Calculated the average score and stored it in the **Average_Score** column.
4. Sorted the dataset in ascending order based on **Average_Score** using the `order()` function.
5. Displayed the first few rows of the sorted dataset.

R Code:

```

1 # Load dataset
2 df <- read.csv("Space_Corrected.csv")
3
4 # Create dummy score columns
5 df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
6 df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75
7
8 # Create Average Score
9 df$Average_Score <- (df$Score1 + df$Score2) / 2
10
11 # Sort data by Average Score (Ascending)
12 sorted_df <- df[order(df$Average_Score), ]
13
14 # Display first 10 rows
15 print(head(sorted_df[c("Score1", "Score2", "Average_Score")], 10))

```

Output & Observations:

The dataset was successfully sorted in ascending order using R. The output matches the Python implementation, correctly displaying the lowest calculated average score (72.5) at the top.

```
R - R 4.6.1 · ~/PycharmProjects/Data_Analysis_Assignments1
> # Load dataset
> df <- read.csv("Space_Corrected.csv")
>
> # Create dummy score columns
> df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
> df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75
>
> # Create Average Score
> df$Average_Score <- (df$Score1 + df$Score2) / 2
>
> # Sort data by Average Score (Ascending)
> sorted_df <- df[order(df$Average_Score), ]
>
> # Display first 10 rows
```

```
R - R 4.6.1 · ~/PycharmProjects/Data_Analysis_Assignments1/
> # Display first 10 rows
> print(head(sorted_df[,c("Score1", "Score2", "Average_Score")], 10))
  Score1 Score2 Average_Score
1      70      75         72.5
337     70      75         72.5
673     70      75         72.5
1009    70      75         72.5
1345    70      75         72.5
1681    70      75         72.5
2017    70      75         72.5
2353    70      75         72.5
2689    70      75         72.5
3025    70      75         72.5
> |
```

5. Conclusion

Both Python and R successfully performed the data sorting operation on the dataset. The computed values and sorted order were consistent across both environments, confirming the accuracy of the implementation in both languages.